Lesson plan · for the teacher

Models that remember too much

AI security · Level 3 · Lesson 2

Time
40 min

Materials
A made-up class register on paper, with names, phone numbers, addresses and notes (one copy per group); Dark pencils or markers; Board and chalk

Learning goals

- Explain memorisation and membership inference.
- Remove identifiers from text before it is used.
- Weigh privacy against usefulness, and ask the right questions.

Activity: Black it out

1. Warm-up (5 min): Show the made-up register. Ask: if a stranger saw this, what could they learn about each pupil? List answers on the board.
2. Teach (8 min): Explain that an AI model trained on text can store rare lines and repeat them. Agree what an identifier is: names, phone and ID numbers, addresses, and rare details that point to one person.
3. Black it out (12 min): Each group blacks out every identifier on its copy with a dark pencil.
4. Swap and check (8 min): Groups swap copies and circle anything the other group missed. Share the best catch with the class.
5. Wrap-up (7 min): Agree one rule for what the school never puts into an AI tool and write it on the board. Then give the exit ticket.

Use only made-up names and numbers, never a real register. Hide two or three rare details (“the only pupil who walks from the river village”) so groups learn that identifiers are not only names and numbers.

Discussion

- Which identifiers were hardest to spot, and why?
- Why can't we just delete the file after a model has learned from it?
- When is it worth making a model less useful to protect privacy?

Exit ticket

- Write three kinds of identifier, and one question to ask before any pupils' data goes to an AI service.

Try it at school tomorrow
Play “Black it out”: give groups a made-up class register with names, phone numbers and notes. Groups black out every identifier by hand, then swap with another group and look for anything the others missed.

Practice questions · for you

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1 **A company offers a free marking assistant if the school uploads every pupil's past essays to train it. What should the school ask first?**

◯ How fast does it mark? ◯ What data does it need, will it train on it, and can it be deleted?

◯ Can it mark in our language?

2 **This redact() should turn "Room 12" into "Room ##", but it returns just "#". Which line has the mistake?**

Circle the line with the mistake. Write it correctly:

```
1  def redact(text):
2      out = ""
3      for c in text:
4          if c.isdigit():
5              out = "#"
6          else:
7              out += c
8      return out
```

3 **Which of these are identifiers you should remove before pupils' text is used to train a model? Choose all that apply.**

☐ A home address ☐ "The only pupil in 5B with twins at home" ☐ A pupil's full name

☐ "Photosynthesis needs light" ☐ A parent's phone number

4 **Deleting the original file also deletes what a model learned from it.**

◯ True ◯ False

5 **Match each defence to what it does.**

Keep personal data out	• • • •	Removes names and numbers before training
Redaction	• • • •	The model never sees it, so it can't leak
Privacy noise	• • • •	Tests whether the model repeats a planted secret
A canary	• • • •	No single example leaves a mark

✎ **What pupils' data does your school hold that should never train a model? If a service asked for it, what questions would you ask first?**

Check your answers

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1 **What data does it need, will it train on it, and can it be deleted?**

Once pupils' essays are inside a model, they are hard to take out and may leak.

2 **Line 5: `out += "#"`**

`out = "#"` throws away everything built so far.

3 **A pupil's full name; A parent's phone number; “The only pupil in 5B with twins at home”; A home address**

Names, numbers and addresses identify a child directly, and a rare detail can point to one child just as well.

4 **False**

What a model learned is stored in its numbers, not in the file.

5 **Keep personal data out — The model never sees it, so it can't leak**
Redaction — Removes names and numbers before training
Privacy noise — No single example leaves a mark
A canary — Tests whether the model repeats a planted secret

Keeping data out prevents the leak; redaction cleans what must go in; privacy noise blurs each example's mark; a canary tests whether leaks happen.